SOWMIYAN S

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PROFESSIONAL SUMMARY

Software engineer with client-facing freelance experience taking AI and web solutions from requirements to production deployment. Built and published LLM, RAG, and multi-agent systems (NPM, PyPI) plus an edge AI computer-vision system, with strong Python, React, and FastAPI fundamentals. Comfortable working directly with customers, debugging end to end, and shipping reliable solutions quickly.

EDUCATION

V.S.B. College of Engineering and Technical Campus, Coimbatore — *Jul 2023 – Present*
B.Tech. Artificial Intelligence and Data Science — **CGPA: 8.5/10**

TECHNICAL SKILLS

- **Programming:** Python, Java, SQL
- **Core CS:** Data Structures & Algorithms, OOP, DBMS
- **Web:** React, JavaScript, HTML/CSS, FastAPI, REST APIs
- **AI / ML:** LLMs, RAG, LangChain, CrewAI, MCP, Fine-Tuning
- **Databases:** MongoDB, MySQL, FAISS
- **Cloud & Deployment:** AWS, Docker
- **Tools:** Git, GitHub Codespaces, Claude Code, Power BI, Android Studio
- **Concepts:** Agentic AI Systems, Prompt Engineering

EXPERIENCE

Freelance Full-Stack Developer & AI Engineer | Self-Employed — *Jan 2023 – Present*

- Worked directly with clients to gather requirements, then designed, built, and deployed custom websites, Discord bots, and AI-powered solutions, owning delivery from first call to production.
- Built websites including a **mechanic shop business site**, a **makeup artist portfolio**, and **college symposium registration portals** handling UI design, backend logic, and hosting end-to-end.
- Developed Discord bots for community automation and created graphic design assets for client brand identity, expanding beyond code to cover full client needs.

Open Source Contributor & Code Reviewer (Volunteer) | ThinAI, Atmega *GitHub* *PlayStore* — *Oct 2026 – Present*

- Redesigned the complete UI of **ThinAI**, a Play Store app by Atmega, improving the look, usability, and overall user experience.
- Contributed to the app's branding, helping shape a more consistent and polished visual identity.
- Appointed **Code Reviewer** by the project owner in recognition of quality open-source contributions, reviewing pull requests for code quality and maintainability.

Python Data Science Intern | AISECT — *Jan 2026 – Feb 2026*

- Studied core machine learning algorithms hands-on and built an **email spam detection model** achieving strong classification accuracy using Python and scikit-learn.
- Explored multi-agent frameworks with mentor guidance, deepening the CrewAI work behind *Crewlyze*, a published NPM package.

PROJECTS

Crewlyze | Multi-Agent Data Analysis Tool (NPM) *GitHub* — *Jan 2026*

- **Tech Stack:** Python, FastAPI, CrewAI, LLM APIs, Pandas, Plotly, Node.js
- Built and published a multi-agent CLI tool on NPM, with 50+ weekly downloads, that turns raw spreadsheets into cleaned datasets, PDF reports, and interactive dashboards.
- Designed a CrewAI pipeline of specialized agents for data cleaning, pattern analysis, and visualization, with a Python backend exposed through an NPM interface.
- Added a natural-language chat assistant that answers questions about the data and generates custom Plotly charts on request.

GuardRAG | Python Package *GitHub* — *Nov 2025*

- **Tech Stack:** Python, FastAPI, LangChain, FastEmbed (ONNX), FAISS, Ollama, SQLite
- Published an open-source Python package for building secure, fully offline document intelligence applications with local LLMs.
- Implemented multi-tier guardrails and customizable security policies to redact sensitive data and detect unsafe inputs.
- Enabled secure one-to-many document access across local devices with real-time responses without internet connectivity.

Cerberus AI | Edge AI / Web Application *GitHub* *Aug 2026*

- **Tech Stack:** Python, YOLOv8, TensorRT, PyTorch, ByteTrack, FastAPI, React, SQLite
- Built a real-time edge AI system to detect missing PPE, including helmets, vests, and masks, from factory camera feeds.
- Developed a remote monitoring dashboard with multi-camera streaming and instant photo alerts for detected safety violations.
- Implemented worker tracking across 12+ live camera feeds with edge inference and false-alarm filtering.

ACHIEVEMENTS & PUBLICATIONS

- Secured **2nd Prize** in an Inter-College Hackathon competing against 20+ teams.
- Published a patent titled **"Smart Dustbin System with Automated Dust Collection, Management, and IoT"**.
- Authored a research paper titled **"A Comprehensive Analysis of Cyber Crimes and Cyber Security Tools"** published in IJCRT.

CERTIFICATIONS

- **Generative AI for Data Science – Microsoft** *Dec 2025*
- **Responsible & Safe AI – NPTEL** *Oct 2025*
- **AWS Cloud Foundations – Infosys Springboard** *Jul 2025*

ACTIVITIES

- **Generative AI Workshop** – Conducted a hands-on workshop on Generative AI and web application development for 50+ students.
- **Tech Content Creator** – Creating AI, development, programming, and technology content on YouTube (@bound-by-code) and Instagram (@bound_by_code).
- **Technical Blogger** – Publishing technical articles and project write-ups on Medium (@sowmiyan_s_).